In [1]:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

data = {
    "Order_ID": [
        1001, 1002, 1003, 1004, 1005, 1006,
        1007, 1008, 1009, 1010, 1011, 1012,
        1013, 1014, 1015, 1016, 1003, 1018,
        1019, 1020, 1021, 1022, 1023, 1024,
        1025, 1026, 1027, 1028, 1029, 1030,
        1031, 1032, 1033, 1034, 1035, 1036,
        1037, 1038, 1039, 1040, 1041, 1042,
        1043, 1044, 1045, 1046, 1047, 1048,
        1049, 1050, 1051, 1052, 1053, 1054
    ],

    "Customer_Name": [
        "  rahul sharma ", "PRIYA REDDY", " Amit Kumar ",
        "sneha rao", None, "  KIRAN  ",
        "ananya reddy", "RAHUL SHARMA", "  arjun reddy ",
        "meena devi", "  ", "Vikram Rao",
        "sai krishna", "  Priya Reddy ", "naveen kumar",
        "divya sharma", " Amit Kumar ", "Rohit Verma",
        "Aarav Mehta", "Sneha Reddy", "Karthik Rao",
        "Ananya Sharma", "Vivek Kumar", "Priya Nair",
        "Rahul Verma", "Meghana Reddy", "Arjun Kumar",
        "Divya Rao", "Sanjay Patel", "Kiran Sharma",
        "Neha Reddy", "Rohit Kumar", "Aarav Reddy",
        "Meena Sharma", "Vikram Patel", "Sneha Rao",
        "Naveen Reddy", "Ananya Kumar", "Sai Teja",
        "Priya Sharma", "Karthik Reddy", "Divya Nair",
        "Rahul Kumar", "Meghana Rao", "Vivek Reddy",
        "Neha Sharma", "Sanjay Kumar", "Kiran Reddy",
        "Arjun Patel", "Meena Reddy", "Rohit Sharma",
        "Sai Krishna", "Ananya Rao", "Naveen Kumar"
    ],

    "Product": [
        " laptop ", "MOBILE", "headphones",
        " TABLET ", "Laptop", " mobile ",
        "Headphones", " laptop", "TABLET",
        " laptop ", None, "MOBILE",
        "headphones", " tablet ", "Laptop",
        "mobile", "headphones", " LAPTOP ",
        "Laptop", "Mobile", "Headphones",
        "Tablet", "Laptop", "Mobile",
        "Headphones", "Tablet", "Laptop",
        "Mobile", "Headphones", "Tablet",
        "Laptop", "Mobile", "Tablet",
        "Headphones", "Laptop", "Mobile",
        "Tablet", "Headphones", "Laptop",
        "Mobile", "Tablet", "Headphones",
        "Laptop", "Mobile", "Tablet",
        "Headphones", "Laptop", "Mobile",
        "Tablet", "Headphones", "Laptop",
        "Mobile", "Tablet", "Headphones"
    ],

    "Quantity": [
        2, 3, 5, 1, 4, "five",
        2, 3, 1, 2, 6, 4,
        np.nan, 2, 3, 5, 5, 2,
        2, 4, 3, 2, 1, 5,
```

```
        4, 3, 2, 6, 3, 2,
        5, 3, 4, 2, 1, 6,
        3, 2, 4, 5, 2, 3,
        4, 2, 5, 3, 6, 2,
        4, 3, 2, 5, 3, 4
    ],

    "Unit_Price": [
        55000, 18000, 1500, 22000, 55000, 18000,
        1500, "unknown", 22000, 55000, 18000, np.nan,
        1500, 22000, 55000, 1558000, 1500, 99999,
        58000, 19000, 1800, 24000, 56000, 17500,
        1600, 23000, 54000, 18500, 1700, 22500,
        57000, 19500, 23500, 1550, 59000, 18500,
        22000, 1650, 56000, 20000, 24500, 1750,
        53000, 19000, 23000, 1600, 61000, 18000,
        22500, 1700, 57500, 19500, 24000, 1550
    ],

    "Order_Date": [
        "2026-01-05", "06/01/2026", "2026-01-07",
        "08-01-2026", "2026-01-09", "10/01/2026",
        "2026-01-11", "12/01/2026", "31/02/2026",
        "2026-01-14", "15/01/2026", "2026-01-16",
        "17/01/2026", "2026-01-18", "19/01/2026",
        "2026-01-20", "2026-01-07", "not-a-date",
        "2026-02-02", "2026-02-05", "2026-02-09",
        "2026-02-12", "2026-02-16", "2026-02-20",
        "2026-02-24", "2026-02-27",
        "2026-03-03", "2026-03-06", "2026-03-10",
        "2026-03-14", "2026-03-18", "2026-03-22",
        "2026-03-26", "2026-03-29",
        "2026-04-02", "2026-04-06", "2026-04-10",
        "2026-04-14", "2026-04-18", "2026-04-22",
        "2026-04-26", "2026-04-29",
        "2026-05-03", "2026-05-07", "2026-05-11",
        "2026-05-15", "2026-05-19", "2026-05-23",
        "2026-05-27", "2026-05-30",
        "2026-06-04", "2026-06-09", "2026-06-15",
        "2026-06-22"
    ],

    "Region": [
        " hyderabad ", "KHAMMAM", "warangal",
        " HYDERABAD", "khammam ", "Warangal",
        "hyderabad", " KHAMMAM ", "warangal",
        "HYDERABAD", None, " khammam",
        "Warangal ", "hyderabad", " KHAMMAM",
        "warangal", "warangal", "HYDERABAD",
        "Hyderabad", "Khammam", "Warangal",
        "Hyderabad", "Khammam", "Warangal",
        "Hyderabad", "Khammam", "Warangal",
        "Hyderabad", "Khammam", "Warangal",
        "Hyderabad", "Khammam", "Warangal",
        "Hyderabad", "Khammam", "Warangal",
        "Hyderabad", "Khammam", "Warangal",
        "Hyderabad", "Khammam", "Warangal",
        "Hyderabad", "Khammam", "Warangal",
        "Hyderabad", "Khammam", "Warangal",
        "Hyderabad", "Khammam", "Warangal",
        "Hyderabad", "Khammam", "Warangal",
    ]
}

df = pd.DataFrame(data)
```

```
df
```

Out[1]:

	Order_ID	Customer_Name	Product	Quantity	Unit_Price	Order_Date	Region
0	1001	rahul sharma	laptop	2	55000	2026-01-05	hyderabad
1	1002	PRIYA REDDY	MOBILE	3	18000	06/01/2026	KHAMMAM
2	1003	Amit Kumar	headphones	5	1500	2026-01-07	warangal
3	1004	sneha rao	TABLET	1	22000	08-01-2026	HYDERABAD
4	1005	NaN	Laptop	4	55000	2026-01-09	khammam
5	1006	KIRAN	mobile	five	18000	10/01/2026	Warangal
6	1007	ananya reddy	Headphones	2	1500	2026-01-11	hyderabad
7	1008	RAHUL SHARMA	laptop	3	unknown	12/01/2026	KHAMMAM
8	1009	arjun reddy	TABLET	1	22000	31/02/2026	warangal
9	1010	meena devi	laptop	2	55000	2026-01-14	HYDERABAD
10	1011		NaN	6	18000	15/01/2026	NaN
11	1012	Vikram Rao	MOBILE	4	NaN	2026-01-16	khammam
12	1013	sai krishna	headphones	NaN	1500	17/01/2026	Warangal
13	1014	Priya Reddy	tablet	2	22000	2026-01-18	hyderabad
14	1015	naveen kumar	Laptop	3	55000	19/01/2026	KHAMMAM
15	1016	divya sharma	mobile	5	1558000	2026-01-20	warangal
16	1003	Amit Kumar	headphones	5	1500	2026-01-07	warangal
17	1018	Rohit Verma	LAPTOP	2	99999	not-a-date	HYDERABAD
18	1019	Aarav Mehta	Laptop	2	58000	2026-02-02	Hyderabad
19	1020	Sneha Reddy	Mobile	4	19000	2026-02-05	Khammam
20	1021	Karthik Rao	Headphones	3	1800	2026-02-09	Warangal
21	1022	Ananya Sharma	Tablet	2	24000	2026-02-12	Hyderabad
22	1023	Vivek Kumar	Laptop	1	56000	2026-02-16	Khammam
23	1024	Priya Nair	Mobile	5	17500	2026-02-20	Warangal
24	1025	Rahul Verma	Headphones	4	1600	2026-02-24	Hyderabad
25	1026	Meghana Reddy	Tablet	3	23000	2026-02-27	Khammam
26	1027	Arjun Kumar	Laptop	2	54000	2026-03-03	Warangal
27	1028	Divya Rao	Mobile	6	18500	2026-03-06	Hyderabad
28	1029	Sanjay Patel	Headphones	3	1700	2026-03-10	Khammam
29	1030	Kiran Sharma	Tablet	2	22500	2026-03-14	Warangal
30	1031	Neha Reddy	Laptop	5	57000	2026-03-18	Hyderabad
31	1032	Rohit Kumar	Mobile	3	19500	2026-03-22	Khammam
32	1033	Aarav Reddy	Tablet	4	23500	2026-03-26	Warangal
33	1034	Meena Sharma	Headphones	2	1550	2026-03-29	Hyderabad
34	1035	Vikram Patel	Laptop	1	59000	2026-04-02	Khammam
35	1036	Sneha Rao	Mobile	6	18500	2026-04-06	Warangal

	Order_ID	Customer_Name	Product	Quantity	Unit_Price	Order_Date	Region
36	1037	Naveen Reddy	Tablet	3	22000	2026-04-10	Hyderabad
37	1038	Ananya Kumar	Headphones	2	1650	2026-04-14	Khammam
38	1039	Sai Teja	Laptop	4	56000	2026-04-18	Warangal
39	1040	Priya Sharma	Mobile	5	20000	2026-04-22	Hyderabad
40	1041	Karthik Reddy	Tablet	2	24500	2026-04-26	Khammam
41	1042	Divya Nair	Headphones	3	1750	2026-04-29	Warangal
42	1043	Rahul Kumar	Laptop	4	53000	2026-05-03	Hyderabad
43	1044	Meghana Rao	Mobile	2	19000	2026-05-07	Khammam
44	1045	Vivek Reddy	Tablet	5	23000	2026-05-11	Warangal
45	1046	Neha Sharma	Headphones	3	1600	2026-05-15	Hyderabad
46	1047	Sanjay Kumar	Laptop	6	61000	2026-05-19	Khammam
47	1048	Kiran Reddy	Mobile	2	18000	2026-05-23	Warangal
48	1049	Arjun Patel	Tablet	4	22500	2026-05-27	Hyderabad
49	1050	Meena Reddy	Headphones	3	1700	2026-05-30	Khammam
50	1051	Rohit Sharma	Laptop	2	57500	2026-06-04	Warangal
51	1052	Sai Krishna	Mobile	5	19500	2026-06-09	Hyderabad
52	1053	Ananya Rao	Tablet	3	24000	2026-06-15	Khammam
53	1054	Naveen Kumar	Headphones	4	1550	2026-06-22	Warangal

Cleaning the Data

In [2]:
```
df["Customer_Name"] = df["Customer_Name"].str.strip().str.title().replace("", np.nan)
df["Product"] = df["Product"].str.strip().str.title()
df["Region"] = df["Region"].str.strip().str.title()

df["Quantity"] = pd.to_numeric(df["Quantity"], errors="coerce")
df["Unit_Price"] = pd.to_numeric(df["Unit_Price"], errors="coerce")

df["Order_Date"] = pd.to_datetime(df["Order_Date"], errors="coerce")

df = df.drop_duplicates(subset="Order_ID", keep="first")
```

In [3]:
```
df.head()
```

Out[3]:

	Order_ID	Customer_Name	Product	Quantity	Unit_Price	Order_Date	Region
0	1001	Rahul Sharma	Laptop	2.0	55000.0	2026-01-05	Hyderabad
1	1002	Priya Reddy	Mobile	3.0	18000.0	NaT	Khammam
2	1003	Amit Kumar	Headphones	5.0	1500.0	2026-01-07	Warangal
3	1004	Sneha Rao	Tablet	1.0	22000.0	NaT	Hyderabad
4	1005	NaN	Laptop	4.0	55000.0	2026-01-09	Khammam

Dataset Inspection

In [4]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
Index: 53 entries, 0 to 53
Data columns (total 7 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Order_ID       53 non-null     int64
 1   Customer_Name  51 non-null     str
 2   Product        52 non-null     str
 3   Quantity       51 non-null     float64
 4   Unit_Price     51 non-null     float64
 5   Order_Date     44 non-null     datetime64[us]
 6   Region         52 non-null     str
dtypes: datetime64[us](1), float64(2), int64(1), str(3)
memory usage: 3.3 KB
```

In [5]:
```
df.shape
```

Out[5]:
```
(53, 7)
```

In [6]:
```
df.columns
```

Out[6]:
```
Index(['Order_ID', 'Customer_Name', 'Product', 'Quantity', 'Unit_Price',
       'Order_Date', 'Region'],
      dtype='str')
```

In [7]:
```
df.dtypes
```

Out[7]:
```
Order_ID                  int64
Customer_Name               str
Product                     str
Quantity                float64
Unit_Price              float64
Order_Date       datetime64[us]
Region                      str
dtype: object
```

In [8]:
```
df.describe()
```

Out[8]:

	Order_ID	Quantity	Unit_Price	Order_Date
count	53.000000	51.000000	5.100000e+01	44
mean	1027.698113	3.235294	5.700782e+04	2026-03-24 03:49:05.454545
min	1001.000000	1.000000	1.500000e+03	2026-01-05 00:00:00
25%	1014.000000	2.000000	1.775000e+04	2026-02-11 06:00:00
50%	1028.000000	3.000000	2.200000e+04	2026-03-24 00:00:00
75%	1041.000000	4.000000	5.450000e+04	2026-05-04 00:00:00
max	1054.000000	6.000000	1.558000e+06	2026-06-22 00:00:00
std	15.814532	1.422508	2.155620e+05	NaN

Missing Values and Duplicates

In [9]:
```
df.isnull()
```

Out[9]:

	Order_ID	Customer_Name	Product	Quantity	Unit_Price	Order_Date	Region
0	False	False	False	False	False	False	False
1	False	False	False	False	False	True	False
2	False	False	False	False	False	False	False
3	False	False	False	False	False	True	False
4	False	True	False	False	False	False	False
5	False	False	False	True	False	True	False
6	False	False	False	False	False	False	False
7	False	False	False	False	True	True	False
8	False	False	False	False	False	True	False
9	False	False	False	False	False	False	False
10	False	True	True	False	False	True	True
11	False	False	False	False	True	False	False
12	False	False	False	True	False	True	False
13	False	False	False	False	False	False	False
14	False	False	False	False	False	True	False
15	False	False	False	False	False	False	False
17	False	False	False	False	False	True	False
18	False	False	False	False	False	False	False
19	False	False	False	False	False	False	False
20	False	False	False	False	False	False	False
21	False	False	False	False	False	False	False
22	False	False	False	False	False	False	False
23	False	False	False	False	False	False	False
24	False	False	False	False	False	False	False
25	False	False	False	False	False	False	False
26	False	False	False	False	False	False	False
27	False	False	False	False	False	False	False
28	False	False	False	False	False	False	False
29	False	False	False	False	False	False	False
30	False	False	False	False	False	False	False
31	False	False	False	False	False	False	False
32	False	False	False	False	False	False	False
33	False	False	False	False	False	False	False
34	False	False	False	False	False	False	False
35	False	False	False	False	False	False	False
36	False	False	False	False	False	False	False

	Order_ID	Customer_Name	Product	Quantity	Unit_Price	Order_Date	Region
37	False	False	False	False	False	False	False
38	False	False	False	False	False	False	False
39	False	False	False	False	False	False	False
40	False	False	False	False	False	False	False
41	False	False	False	False	False	False	False
42	False	False	False	False	False	False	False
43	False	False	False	False	False	False	False
44	False	False	False	False	False	False	False
45	False	False	False	False	False	False	False
46	False	False	False	False	False	False	False
47	False	False	False	False	False	False	False
48	False	False	False	False	False	False	False
49	False	False	False	False	False	False	False
50	False	False	False	False	False	False	False
51	False	False	False	False	False	False	False
52	False	False	False	False	False	False	False
53	False	False	False	False	False	False	False

In [10]:
```
df.isnull().sum()
```

Out[10]:
```
Order_ID         0
Customer_Name    2
Product          1
Quantity         2
Unit_Price       2
Order_Date       9
Region           1
dtype: int64
```

In [11]:
```
df.isnull().sum()
```

Out[11]:
```
Order_ID         0
Customer_Name    2
Product          1
Quantity         2
Unit_Price       2
Order_Date       9
Region           1
dtype: int64
```

In [44]:
```
df.duplicated()
```

Out[44]:
```
0     False
1     False
2     False
3     False
4     False
5     False
6     False
7     False
8     False
9     False
10    False
11    False
12    False
13    False
14    False
15    False
17    False
18    False
19    False
20    False
21    False
22    False
23    False
24    False
25    False
26    False
27    False
28    False
29    False
30    False
31    False
32    False
33    False
34    False
35    False
36    False
37    False
38    False
39    False
40    False
41    False
42    False
43    False
44    False
45    False
46    False
47    False
48    False
49    False
50    False
51    False
52    False
53    False
dtype: bool
```

In [13]:
```
df["Order_ID"].duplicated().sum()
```

Out[13]:
```
np.int64(0)
```

In [14]:
```
df["Order_ID"].duplicated().sum()
```

Out[14]:
```
np.int64(0)
```

Creating the revenue

In [15]:
```
df["Revenue"] = df["Quantity"] * df["Unit_Price"]
```

In [45]:
```
df[["Order_ID", "Product", "Quantity", "Unit_Price", "Revenue"]].head()
```

Out[45]:

	Order_ID	Product	Quantity	Unit_Price	Revenue
0	1001	Laptop	2.0	55000.0	110000.0
1	1002	Mobile	3.0	18000.0	54000.0
2	1003	Headphones	5.0	1500.0	7500.0
3	1004	Tablet	1.0	22000.0	22000.0
4	1005	Laptop	4.0	55000.0	220000.0

In [17]:
```
df["Revenue"].describe()
```

Out[17]:
```
count    4.900000e+01
mean     2.409112e+05
std      1.103668e+06
min      3.000000e+03
25%      2.200000e+04
50%      6.900000e+04
75%      1.110000e+05
max      7.790000e+06
Name: Revenue, dtype: float64
```

In [18]:
```
df["Revenue"].sum()
```

Out[18]:
```
np.float64(11804648.0)
```

In [19]:
```
meanRev=df["Revenue"].mean()

meanRev
```

Out[19]:
```
np.float64(240911.18367346938)
```

In [20]:
```
df["Revenue"].max()
```

Out[20]:
```
np.float64(7790000.0)
```

In [21]:
```
df["Revenue"].min()
```

Out[21]:
```
np.float64(3000.0)
```

Unique Values and Counts

In [22]:
```
df["Product"].unique()
```

Out[22]:
```
<StringArray>
['Laptop', 'Mobile', 'Headphones', 'Tablet', nan]
Length: 5, dtype: str
```

In [23]:
```
df["Product"].unique().count()
```

Out[23]:
```
np.int64(4)
```

In [24]:
```
df["Product"].nunique()
```

Out[24]:
```
4
```

In [25]:
```
df["Region"].unique()
```

Out[25]:
```
<StringArray>
['Hyderabad', 'Khammam', 'Warangal', nan]
Length: 4, dtype: str
```

In [26]:
```
df["Region"].nunique()
```

Out[26]:
```
3
```

In [27]:
```
df["Region"].value_counts()
```

Out[27]:
```
Region
Hyderabad    18
Khammam      17
Warangal     17
Name: count, dtype: int64
```

Revenue by Region

In [28]:
```
revenue_by_region = df.groupby("Region")["Revenue"].sum().sort_values(ascending=False)

revenue_by_region
```

Out[28]:
```
Region
Warangal     8771850.0
Hyderabad    1628798.0
Khammam      1296000.0
Name: Revenue, dtype: float64
```

In [29]:
```
df.groupby("Region")["Revenue"].mean().sort_values(ascending=False)
```

Out[29]:
```
Region
Warangal     584790.000000
Hyderabad     90488.777778
Khammam       86400.000000
Name: Revenue, dtype: float64
```

In [30]:
```
df.groupby("Region")["Order_ID"].count().sort_values(ascending=False)
```

Out[30]:
```
Region
Hyderabad    18
Khammam      17
Warangal     17
Name: Order_ID, dtype: int64
```

In [31]:
```
df.groupby(["Region", "Product"])["Revenue"].sum().sort_values(ascending=False)
```

Out[31]:
```
Region     Product
Warangal   Mobile        8024500.0
Hyderabad  Laptop        1032998.0
Khammam    Laptop         866000.0
Warangal   Laptop         447000.0
Hyderabad  Mobile         308500.0
Warangal   Tablet         276000.0
Hyderabad  Tablet         270000.0
Khammam    Mobile         226500.0
           Tablet         190000.0
Warangal   Headphones      24350.0
Hyderabad  Headphones      17300.0
Khammam    Headphones      13500.0
Name: Revenue, dtype: float64
```

GroupBy and Aggregation

In [32]:
```
df.groupby("Product")["Revenue"].agg(["sum", "mean", "min", "max", "count"])
```

Out[32]:

Product	sum	mean	min	max	count
Headphones	55150.0	5013.636364	3000.0	7500.0	11
Laptop	2345998.0	167571.285714	56000.0	366000.0	14
Mobile	8559500.0	778136.363636	36000.0	7790000.0	11
Tablet	736000.0	61333.333333	22000.0	115000.0	12

In [33]:
```
df.groupby("Product").agg(
    Total_Quantity=("Quantity", "sum"),
    Average_Quantity=("Quantity", "mean"),
    Total_Revenue=("Revenue", "sum"),
    Average_Revenue=("Revenue", "mean"),
    Number_Of_Orders=("Order_ID", "count")
)
```

Out[33]:

Product	Total_Quantity	Average_Quantity	Total_Revenue	Average_Revenue	Number_Of_Orders
Headphones	34.0	3.090909	55150.0	5013.636364	12
Laptop	43.0	2.866667	2345998.0	167571.285714	15
Mobile	50.0	4.166667	8559500.0	778136.363636	13
Tablet	32.0	2.666667	736000.0	61333.333333	12

In [34]:
```
df.groupby(["Region", "Product"]).agg(
    Total_Revenue=("Revenue", "sum"),
    Total_Quantity=("Quantity", "sum"),
    Average_Price=("Unit_Price", "mean"),
    Order_Count=("Order_ID", "count")
).sort_values("Total_Revenue", ascending=False)
```

Out[34]:

Region	Product	Total_Revenue	Total_Quantity	Average_Price	Order_Count
Warangal	Mobile	8024500.0	18.0	326000.000000	5
Hyderabad	Laptop	1032998.0	17.0	62999.833333	6
Khammam	Laptop	866000.0	18.0	57200.000000	6
Warangal	Laptop	447000.0	8.0	55833.333333	3
Hyderabad	Mobile	308500.0	16.0	19333.333333	3
Warangal	Tablet	276000.0	12.0	22750.000000	4
Hyderabad	Tablet	270000.0	12.0	22500.000000	5
Khammam	Mobile	226500.0	16.0	18875.000000	5
	Tablet	190000.0	8.0	23833.333333	3
Warangal	Headphones	24350.0	15.0	1620.000000	5
Hyderabad	Headphones	17300.0	11.0	1562.500000	4
Khammam	Headphones	13500.0	8.0	1683.333333	3

In [35]:
```
df.groupby("Region")[["Quantity", "Unit_Price", "Revenue"]].agg(["sum", "mean", "max"])
```

Out[35]:

	Quantity			Unit_Price			Rev	
	sum	mean	max	sum	mean	max	sum	mean
Region								
Hyderabad	56.0	3.111111	6.0	554749.0	30819.388889	99999.0	1628798.0	90488.777778	285
Khammam	50.0	2.941176	6.0	438050.0	29203.333333	61000.0	1296000.0	86400.000000	366
Warangal	53.0	3.533333	6.0	1896600.0	111564.705882	1558000.0	8771850.0	584790.000000	7790

Numeric Columns and Correlation

In [36]:
```
numeric_df = df.select_dtypes(include="number")

numeric_df
```

Out[36]:

	Order_ID	Quantity	Unit_Price	Revenue
0	1001	2.0	55000.0	110000.0
1	1002	3.0	18000.0	54000.0
2	1003	5.0	1500.0	7500.0
3	1004	1.0	22000.0	22000.0
4	1005	4.0	55000.0	220000.0
5	1006	NaN	18000.0	NaN
6	1007	2.0	1500.0	3000.0
7	1008	3.0	NaN	NaN
8	1009	1.0	22000.0	22000.0
9	1010	2.0	55000.0	110000.0
10	1011	6.0	18000.0	108000.0
11	1012	4.0	NaN	NaN
12	1013	NaN	1500.0	NaN
13	1014	2.0	22000.0	44000.0
14	1015	3.0	55000.0	165000.0
15	1016	5.0	1558000.0	7790000.0
17	1018	2.0	99999.0	199998.0
18	1019	2.0	58000.0	116000.0
19	1020	4.0	19000.0	76000.0
20	1021	3.0	1800.0	5400.0
21	1022	2.0	24000.0	48000.0
22	1023	1.0	56000.0	56000.0
23	1024	5.0	17500.0	87500.0
24	1025	4.0	1600.0	6400.0
25	1026	3.0	23000.0	69000.0
26	1027	2.0	54000.0	108000.0
27	1028	6.0	18500.0	111000.0
28	1029	3.0	1700.0	5100.0
29	1030	2.0	22500.0	45000.0
30	1031	5.0	57000.0	285000.0
31	1032	3.0	19500.0	58500.0
32	1033	4.0	23500.0	94000.0
33	1034	2.0	1550.0	3100.0
34	1035	1.0	59000.0	59000.0
35	1036	6.0	18500.0	111000.0
36	1037	3.0	22000.0	66000.0

	Order_ID	Quantity	Unit_Price	Revenue
37	1038	2.0	1650.0	3300.0
38	1039	4.0	56000.0	224000.0
39	1040	5.0	20000.0	100000.0
40	1041	2.0	24500.0	49000.0
41	1042	3.0	1750.0	5250.0
42	1043	4.0	53000.0	212000.0
43	1044	2.0	19000.0	38000.0
44	1045	5.0	23000.0	115000.0
45	1046	3.0	1600.0	4800.0
46	1047	6.0	61000.0	366000.0
47	1048	2.0	18000.0	36000.0
48	1049	4.0	22500.0	90000.0
49	1050	3.0	1700.0	5100.0
50	1051	2.0	57500.0	115000.0
51	1052	5.0	19500.0	97500.0
52	1053	3.0	24000.0	72000.0
53	1054	4.0	1550.0	6200.0

In [37]:
```
numeric_df.columns
```

Out[37]:
```
Index(['Order_ID', 'Quantity', 'Unit_Price', 'Revenue'], dtype='str')
```

In [38]:
```
df.select_dtypes(include="number").columns.tolist()
```

Out[38]:
```
['Order_ID', 'Quantity', 'Unit_Price', 'Revenue']
```

In [39]:
```
numeric_df.describe()
```

Out[39]:

	Order_ID	Quantity	Unit_Price	Revenue
count	53.000000	51.000000	5.100000e+01	4.900000e+01
mean	1027.698113	3.235294	5.700782e+04	2.409112e+05
std	15.814532	1.422508	2.155620e+05	1.103668e+06
min	1001.000000	1.000000	1.500000e+03	3.000000e+03
25%	1014.000000	2.000000	1.775000e+04	2.200000e+04
50%	1028.000000	3.000000	2.200000e+04	6.900000e+04
75%	1041.000000	4.000000	5.450000e+04	1.110000e+05
max	1054.000000	6.000000	1.558000e+06	7.790000e+06

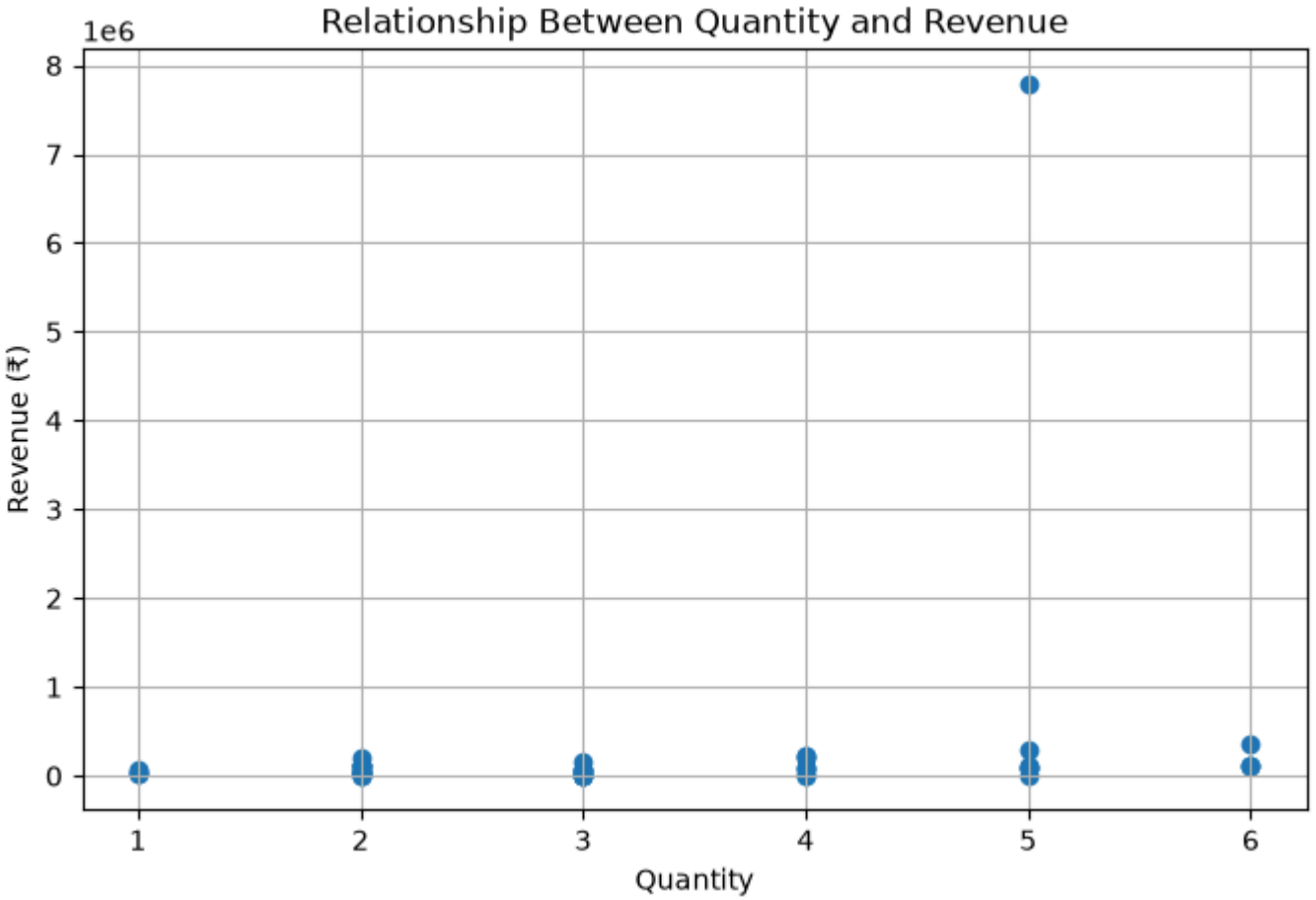
In [40]:
```
numeric_df.corr()
```

Out[40]:

	Order_ID	Quantity	Unit_Price	Revenue
Order_ID	1.000000	0.162354	-0.122319	-0.120159
Quantity	0.162354	1.000000	0.163268	0.208501
Unit_Price	-0.122319	0.163268	1.000000	0.997668
Revenue	-0.120159	0.208501	0.997668	1.000000

Scatter Plot

In [46]:



Monthly Revenue

In [41]:
```
df["Order_Month"] = df["Order_Date"].dt.to_period("M")
```

In [42]:
```
monthly_revenue = df.groupby("Order_Month")["Revenue"].sum().sort_index()

monthly_revenue
```

Out[42]:
```
Order_Month
2026-01    8284500.0
2026-02     464300.0
2026-03     709700.0
2026-04     617550.0
2026-05     866900.0
2026-06     290700.0
Freq: M, Name: Revenue, dtype: float64
```

Monthly Summary

In [47]:
```
monthly_summary = df.groupby("Order_Month").agg(
    Total_Revenue=("Revenue", "sum"),
    Average_Revenue=("Revenue", "mean"),
    Total_Quantity=("Quantity", "sum"),
    Order_Count=("Order_ID", "count")
)

monthly_summary
```

Out[47]:

Order_Month	Total_Revenue	Average_Revenue	Total_Quantity	Order_Count
2026-01	8284500.0	1183500.00	26.0	8
2026-02	464300.0	58037.50	24.0	8
2026-03	709700.0	88712.50	27.0	8
2026-04	617550.0	77193.75	26.0	8
2026-05	866900.0	108362.50	29.0	8
2026-06	290700.0	72675.00	14.0	4

Product Summary

In [48]:
```
product_summary = df.groupby("Product").agg(
    Total_Orders=("Order_ID", "count"),
    Total_Quantity=("Quantity", "sum"),
    Average_Price=("Unit_Price", "mean"),
    Total_Revenue=("Revenue", "sum"),
    Average_Revenue=("Revenue", "mean")
).sort_values("Total_Revenue", ascending=False)

product_summary
```

Out[48]:

Product	Total_Orders	Total_Quantity	Average_Price	Total_Revenue	Average_Revenue
Mobile	13	50.0	146958.333333	8559500.0	778136.363636
Laptop	15	43.0	59392.785714	2345998.0	167571.285714
Tablet	12	32.0	22916.666667	736000.0	61333.333333
Headphones	12	34.0	1616.666667	55150.0	5013.636364

Region Summary

In [50]:
```
region_summary = df.groupby("Region").agg(
    Total_Orders=("Order_ID", "count"),
    Total_Quantity=("Quantity", "sum"),
    Total_Revenue=("Revenue", "sum"),
    Average_Revenue=("Revenue", "mean")
).sort_values("Total_Revenue", ascending=False)

region_summary
```

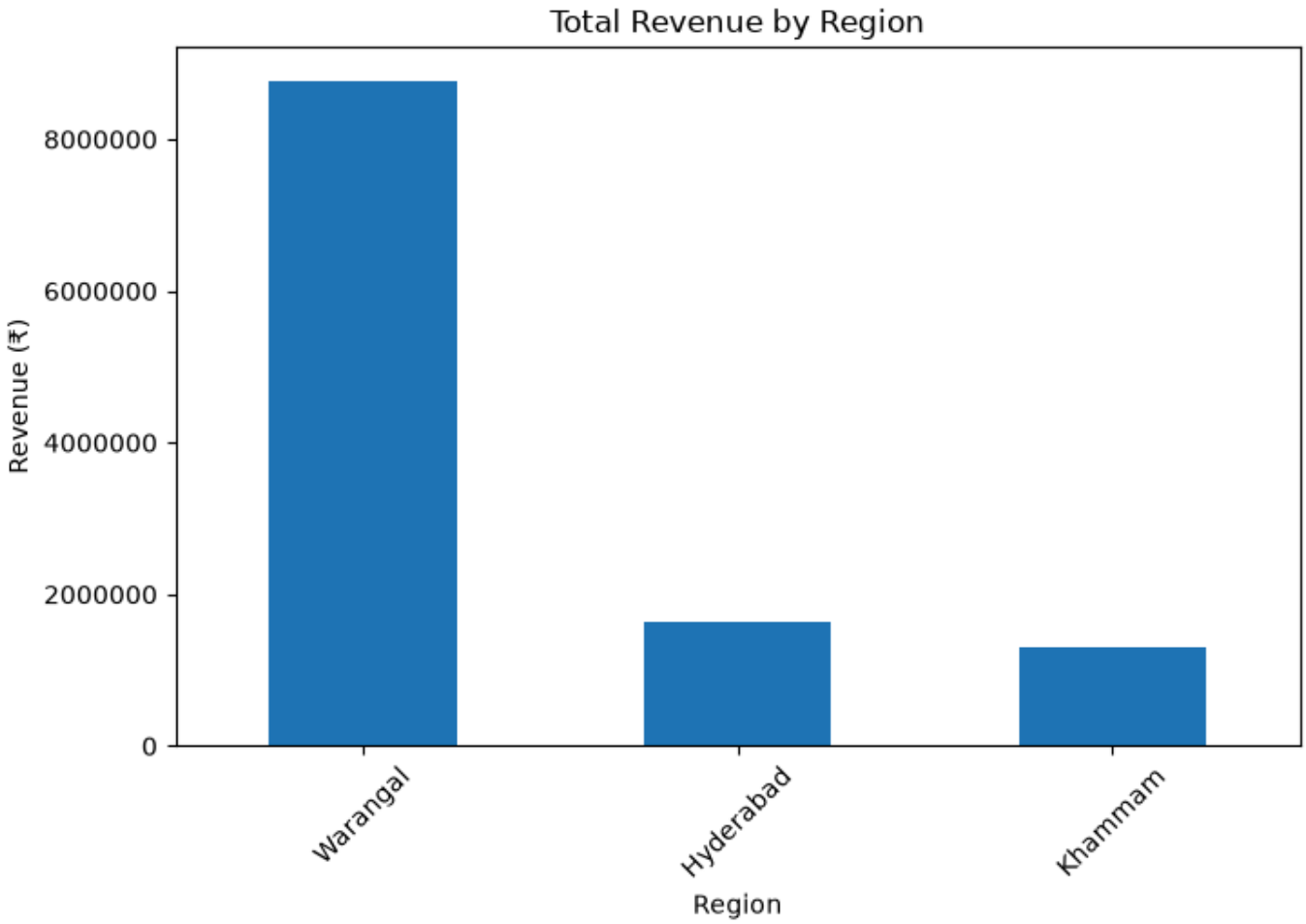
Out[50]:

Region	Total_Orders	Total_Quantity	Total_Revenue	Average_Revenue
Warangal	17	53.0	8771850.0	584790.000000
Hyderabad	18	56.0	1628798.0	90488.777778
Khammam	17	50.0	1296000.0	86400.000000

Revenue by Region Plot

In [57]:
```
revenue_by_region = df.groupby("Region")["Revenue"].sum().sort_values(ascending=False)

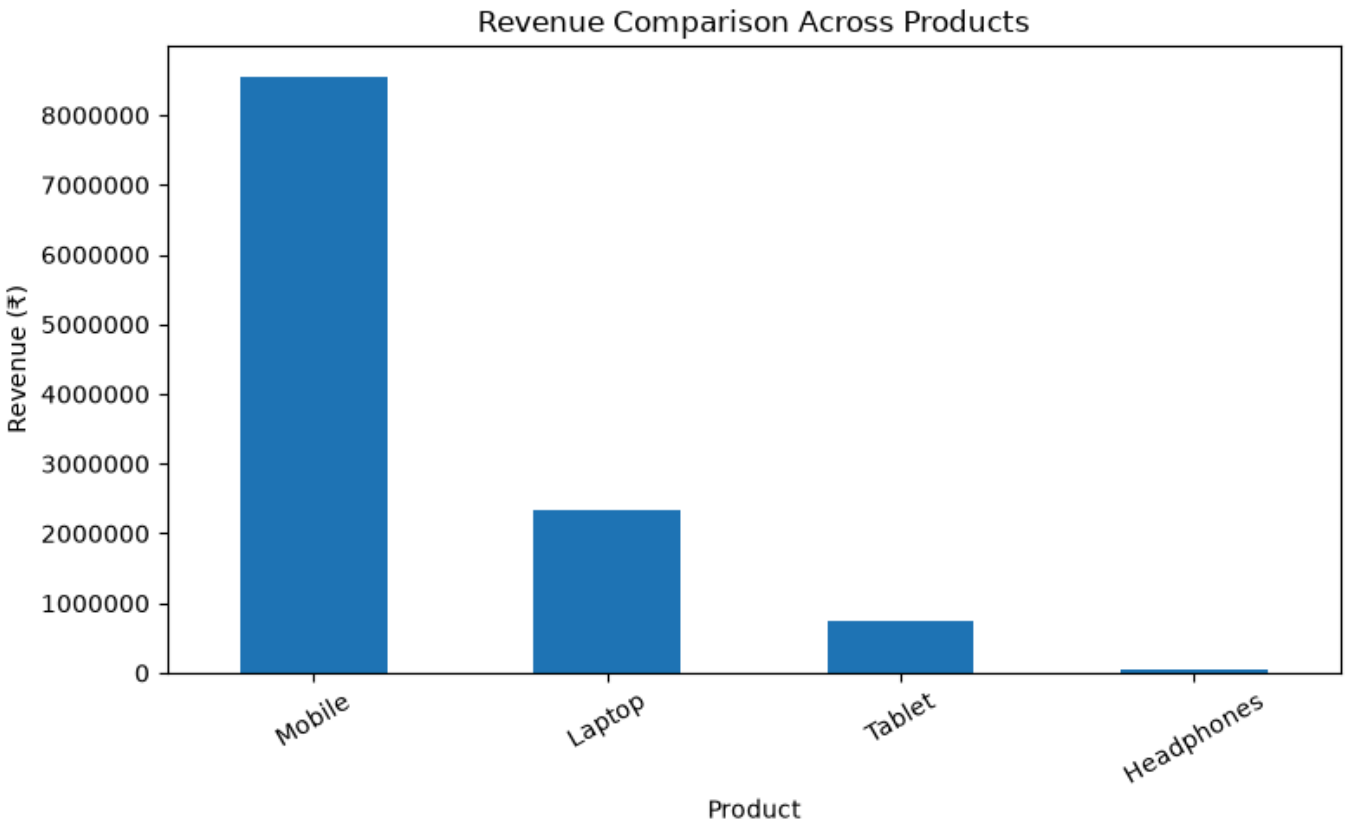
plt.figure(figsize=(8, 5))
revenue_by_region.plot(kind="bar")
plt.ticklabel_format(style="plain", axis="y")
plt.title("Total Revenue by Region")
plt.xlabel("Region")
plt.ylabel("Revenue (₹)")
plt.xticks(rotation=45)
plt.show()
```



Product Revenue Plot

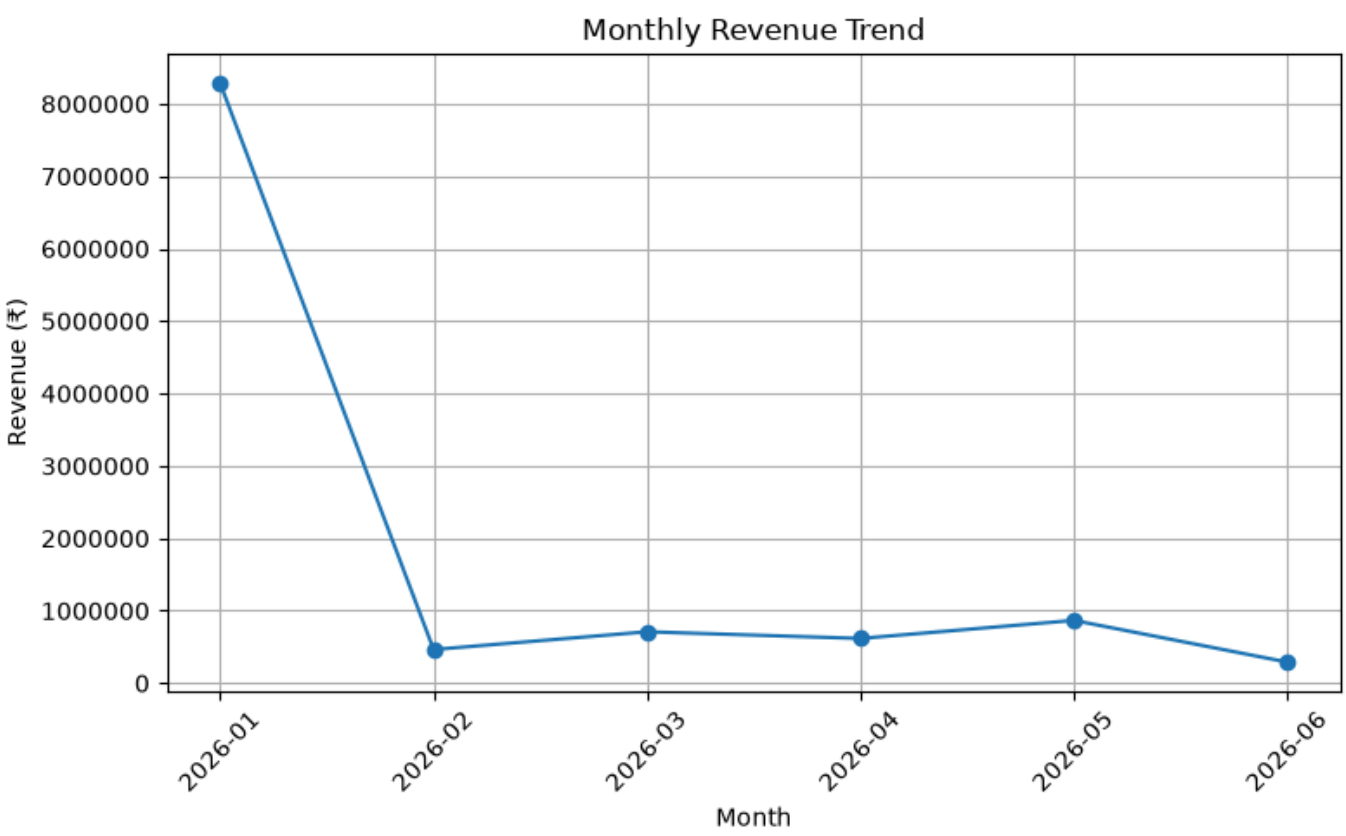
In [58]:
```
plt.figure(figsize=(8, 5))
product_summary["Total_Revenue"].plot(kind="bar")
plt.title("Revenue Comparison Across Products")
plt.xlabel("Product")
plt.ylabel("Revenue (₹)")
plt.ticklabel_format(style="plain", axis="y")
plt.xticks(rotation=30)
```

```
plt.tight_layout()
plt.show()
```



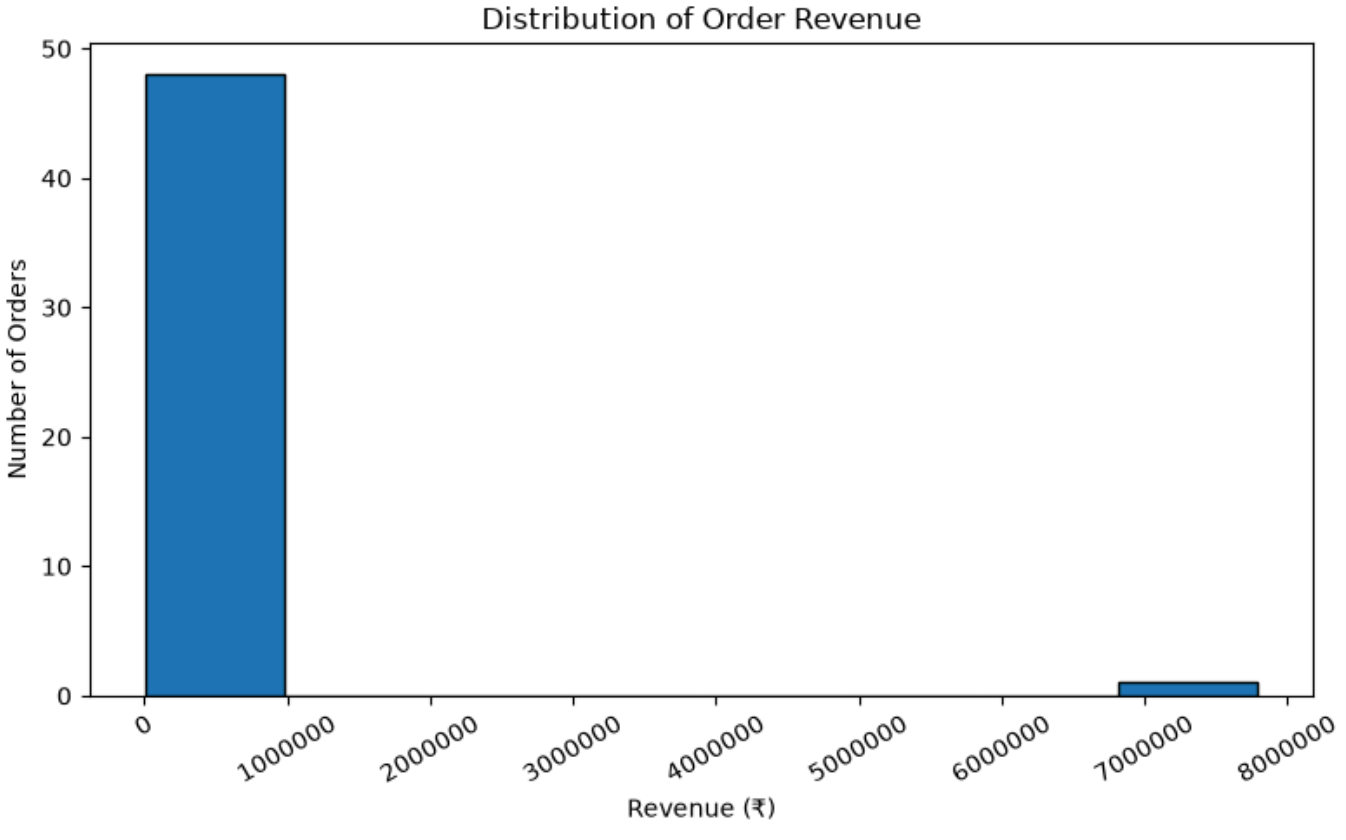
Monthly Revenue Line Plot

In [60]:
```
plt.figure(figsize=(8, 5))
plt.plot(monthly_revenue.index.astype(str), monthly_revenue.values, marker="o")
plt.title("Monthly Revenue Trend")
plt.xlabel("Month")
plt.ylabel("Revenue (₹)")
plt.ticklabel_format(style="plain", axis="y")
plt.xticks(range(len(monthly_revenue)), monthly_revenue.index.astype(str), rotation=45)
plt.grid(True)
plt.tight_layout()
plt.show()
```



Revenue Histogram

In [65]:
```
plt.figure(figsize=(8, 5))
plt.hist(df["Revenue"].dropna(), bins=8, edgecolor="black")
plt.title("Distribution of Order Revenue")
plt.xlabel("Revenue (₹)")
plt.ylabel("Number of Orders")
plt.ticklabel_format(style="plain", axis="y")
plt.ticklabel_format(style="plain", axis="x")
plt.xticks(rotation=30)
plt.tight_layout()
plt.show()
```



Highest Revenue Orders

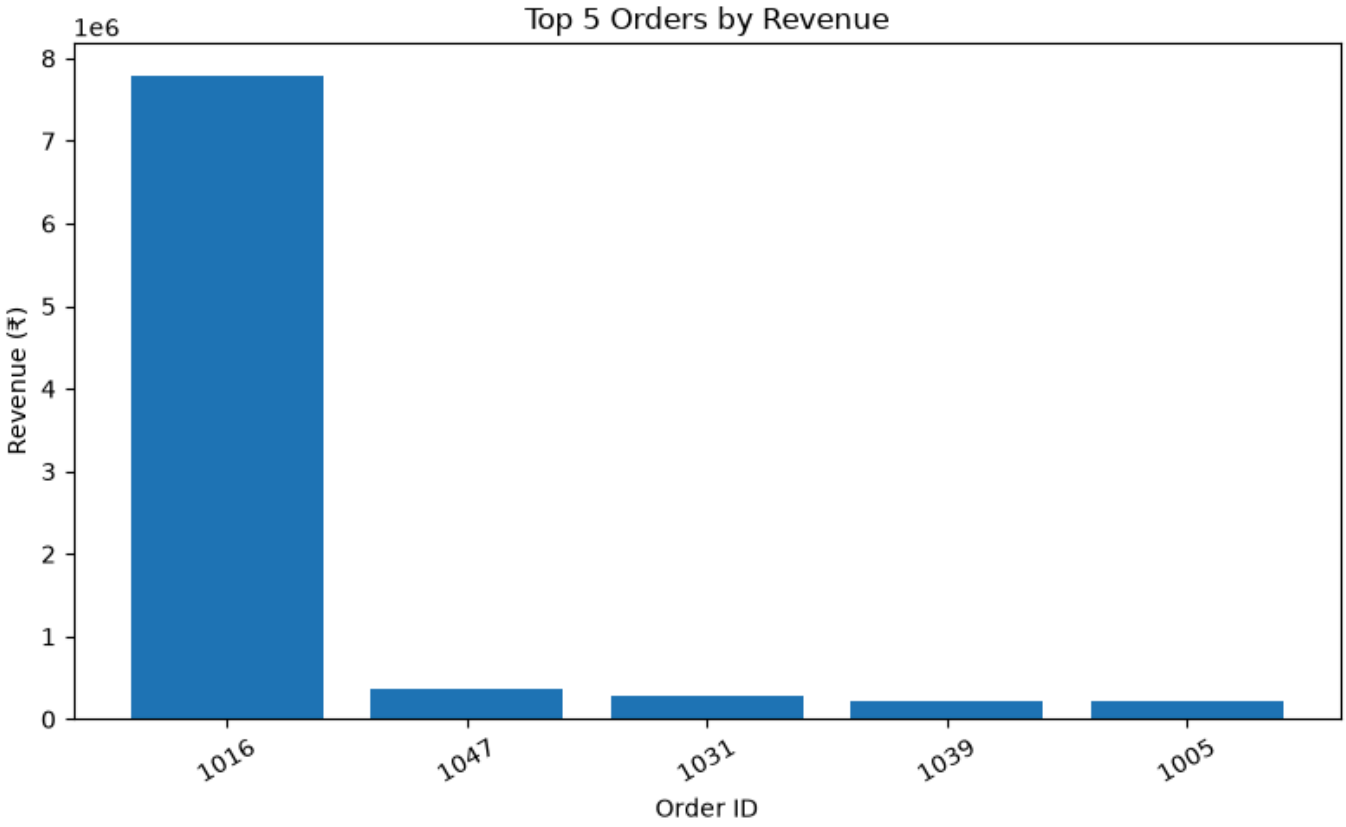
In [66]:
```
highest_orders = df.nlargest(5, "Revenue")[[
    "Order_ID", "Customer_Name", "Product",
    "Quantity", "Unit_Price", "Revenue", "Region"
]]

highest_orders
```

Out[66]:

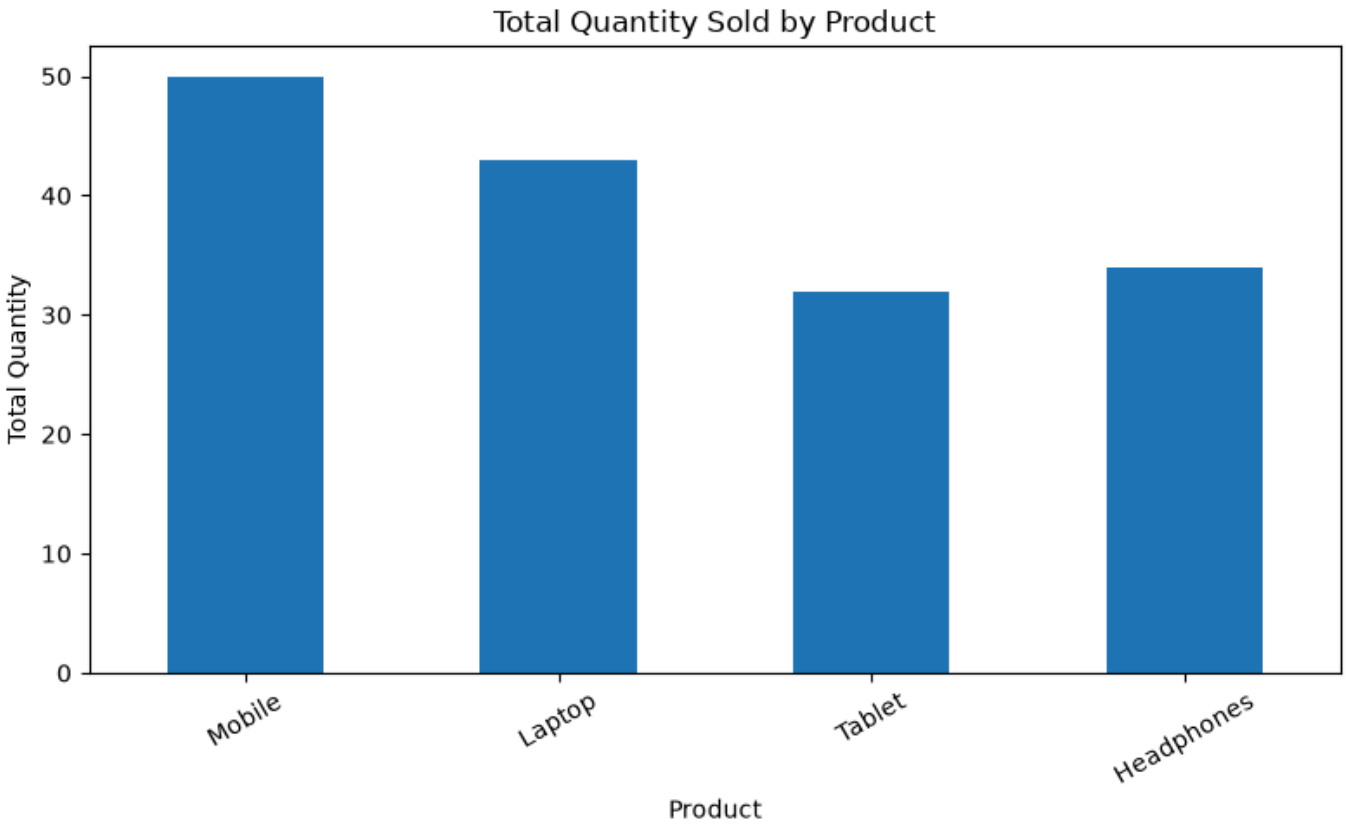
	Order_ID	Customer_Name	Product	Quantity	Unit_Price	Revenue	Region
15	1016	Divya Sharma	Mobile	5.0	1558000.0	7790000.0	Warangal
46	1047	Sanjay Kumar	Laptop	6.0	61000.0	366000.0	Khammam
30	1031	Neha Reddy	Laptop	5.0	57000.0	285000.0	Hyderabad
38	1039	Sai Teja	Laptop	4.0	56000.0	224000.0	Warangal
4	1005	NaN	Laptop	4.0	55000.0	220000.0	Khammam

In [68]:
```
plt.figure(figsize=(8, 5))
plt.bar(highest_orders["Order_ID"].astype(str), highest_orders["Revenue"])
plt.title("Top 5 Orders by Revenue")
plt.xlabel("Order ID")
plt.ylabel("Revenue (₹)")
plt.xticks(rotation=30)
plt.tight_layout()
plt.show()
```



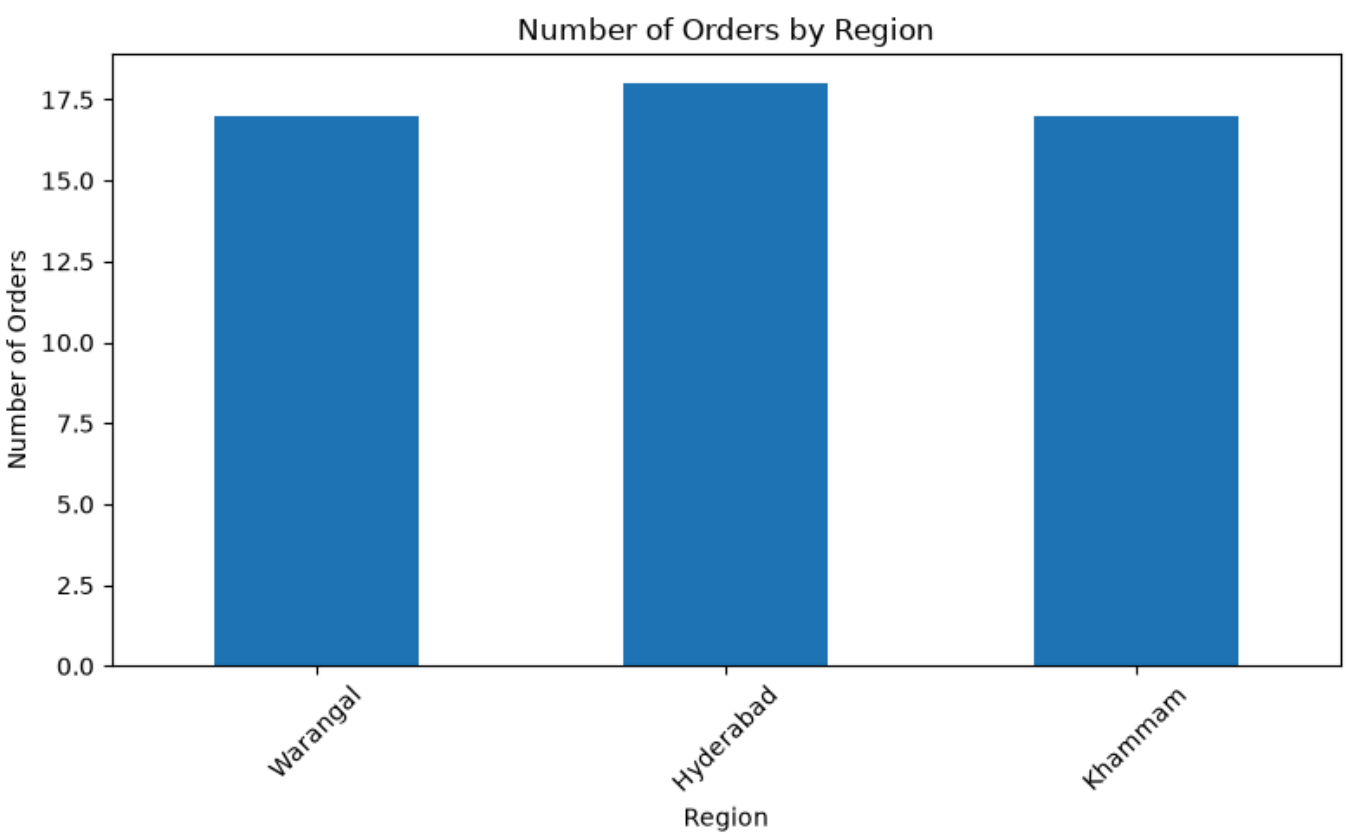
In [70]:
```
plt.figure(figsize=(8, 5))
product_summary["Total_Quantity"].plot(kind="bar")
plt.title("Total Quantity Sold by Product")
plt.xlabel("Product")
plt.ylabel("Total Quantity")
plt.xticks(rotation=30)
```

```
plt.tight_layout()
plt.show()
```



Number of Orders by Region

In [72]:
```
plt.figure(figsize=(8, 5))
region_summary["Total_Orders"].plot(kind="bar")
plt.title("Number of Orders by Region")
plt.xlabel("Region")
plt.ylabel("Number of Orders")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```


Number of Orders by Region
Number of Orders
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
Warangal
Hyderabad
Khammam
Region